4. What are the three different types of Collagen?

5. What are the non-bonded interactions in tertiary structure of a protein?

Answer the following in detail

1. Explain the structure of Globular proteins with an example.

2. Write about the hierarchical levels of protein structure.

3. Distinguish between α helix and β pleated sheets.

4. What is a peptide bond and what are its important properties?

5. Write any five physical and chemical properties of proteins.

SUMMARY

Proteins are important class of macromolecules which are the work horses of the cell. They perform important functions like catalysis of metabolic reactions, defence against foreign bodies, structural support for the cell, transport of other molecules etc. Proteins are polymers of twenty different amino acids. There are 20 amino acids which can be constituents of proteins. Based on the properties of their side chain chemical groups, amino acids can be classified into hydrophobic, polar, acidic and basic amino acids. The polymers of these 20 amino acids are known as polypeptide chains. One or more polypeptide chains can form a protein molecule. These polypeptide chains can fold to obtain a specific three dimensional structure and obtain specific function. Proteins structures can be described in a hierarchical order known as primary, secondary, tertiary and quaternary structures. Proper folding of a polypeptide chain to a proper three dimensional structure is important for a protein to achieve its biological function. So misfolding of proteins often leads to several diseases.

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